DEPI PROJECT

NUVANTA

AI-Powered Enterprise Financial & Operations Management System

Requirements Gathering

Document	Requirements Gathering
Project	NUVANTA
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1. Stakeholder Analysis

1.1 Approach

Stakeholders are the people and organisations who use NUVANTA, approve its work, or are affected by it. Because NUVANTA is a graduation project, the needs below are derived from the NUVANTA Blueprint, the Project Proposal and general knowledge of finance and procurement processes, and are reviewed with the instructor. Where real users are available (for example classmates or a company contact), their feedback is added through short interviews and usability tests (risk R13).

1.2 Stakeholder Register

Stakeholder	Key Needs and Expectations	Influence	Interest	Strategy
Company Owner / CEO	Executive view of budget, cash, spending and forecasts; approves large operations; sets financial policy	High	High	Manage closely
Finance Manager / CFO	Creates budgets, controls limits, reviews cash flow and commitments, approves within delegation, needs trustworthy reports	High	High	Manage closely
Accountant	Records invoices, receipts and payments; reliable ledger; audit trail; cannot approve own entries	Medium	High	Keep informed
Department Manager	Sees department budget (planned, committed, spent, available); approves requests within a limit; tracks consumption	Medium	High	Keep informed
Procurement Officer	Creates purchase orders, manages suppliers, compares prices, tracks receipt; has no authority to release money	Medium	High	Keep informed
Inventory / Warehouse Officer	Receives, issues and transfers stock; sees stock levels and low-stock alerts; limited to assigned warehouses	Medium	Medium	Keep informed
Employee / Requester	Simple request form, clear status, notifications; sees only own data	Low	High	Keep informed
Auditor (read-only)	Reviews records, changes and approvals without creating, editing or deleting anything	Medium	Medium	Keep satisfied
External Supplier	Receives purchase orders, uploads invoices, follows receipt and payment; never sees internal budgets	Low	Medium	Monitor
Platform Administrator	Manages platform settings and operation without automatic access to company financial data	Medium	Medium	Keep satisfied
Payroll/HR, Production and Sales Officers (Phase 2)	Employee and salary data, production costing, customers and sales; sensitive data only with explicit permission	Low	Medium	Monitor
Instructor and DEPI evaluators	Clear scope, quality documentation, a working end-to-end demo, on-time deliverables	High	High	Manage closely
Development team	A scope that can be built and tested in the available time; clear, testable requirements	High	High	Manage closely
Banks and payment providers (external)	Official integration only; legal and Central Bank of Egypt requirements respected	Low	Low	Monitor

1.3 Influence / Interest Grid

	Low interest	High interest
High influence	**Keep satisfied:** Auditor, Platform Administrator	**Manage closely:** CEO, CFO, Instructor and DEPI, Development team
Low influence	**Monitor:** External Supplier, Phase 2 officers, Banks and payment providers	**Keep informed:** Accountant, Department Manager, Procurement, Warehouse, Employee

Influence is judged by how much a group can change scope or accept the result; Medium roles are placed by their main effect on daily use. This grid is a planning aid and can be adjusted as the SRD is finalised.

1.4 Main Problems the Stakeholders Share

- Budgets exceeded without anyone noticing in time.
- Purchases approved without checking available budget or cash.
- No clear difference between planned, committed, spent and cash actually available.
- Weak audit trail: unclear who changed what and when.
- Late and manual decisions because forecasting and analysis are not built in.

2. User Stories & Use Cases

2.1 User Stories

Priority uses MoSCoW: **M** Must (needed for the Phase 1 demo), **S** Should (important, built if time allows), **C** Could (nice to have).

ID	As a ...	I want to ...	So that ...	Pri.
US-01	Administrator	create users and assign roles and departments	each person can access only what their job needs	M
US-02	User	log in securely and manage my password and sessions	my account stays protected	M
US-03	CFO	create a budget for a period and distribute it to departments, cost centers and categories	company money is planned before it is spent	M
US-04	CFO	set limits and receive an alert when spending reaches a set percentage	overruns are noticed early	S
US-05	CEO	open a dashboard with total budget, cash, spent, committed and available amounts	I understand the financial position at a glance	M
US-06	Employee	submit a purchase request with item, quantity, estimated price and justification	I can get the supplies I need	M
US-07	Employee	follow the status of my requests and receive notifications	I do not have to ask anyone for updates	M
US-08	Department Manager	review and approve requests within my financial limit	routine purchases are not delayed	M
US-09	Department Manager	see my department budget: planned, committed, spent and available	I can control consumption	M
US-10	CFO	approve high-value requests under approval rules, never my own transactions	segregation of duties (Maker–Checker) is respected	M
US-11	Finance Manager	have approved requests reserve budget automatically	available budget is never overstated	M
US-12	Procurement Officer	turn an approved request into a purchase order for a chosen supplier	buying follows the approved request	M
US-13	Warehouse Officer	record goods receipt against a purchase order and check quantities	stock and payables match what actually arrived	M
US-14	Accountant	register a supplier invoice and match it to the order and receipt	I pay only for what was ordered and received	M
US-15	Accountant	record a payment (sandbox or external transfer with reference and receipt)	the liability is settled and traceable	M
US-16	Accountant	be sure a failed payment stays unpaid and a repeated confirmation is not counted twice	no duplicate or false payment is recorded	M
US-17	Warehouse Officer	see stock levels and low-stock alerts	I can reorder before items run out	S
US-18	Accountant	see ledger entries created from each transaction	the books are consistent and reviewable	M
US-19	Auditor	read an audit log showing who changed what, when, and the old and new values	I can verify and trace changes	M
US-20	Supplier	view purchase orders, upload invoices and see payment status	I do not need calls or emails to follow up	C
US-21	CFO	see a cash-flow forecast for the coming weeks or months	I can act before a shortage	S
US-22	CFO	be warned when a department is likely to exceed its budget	I can intervene in time	S

ID	As a ...	I want to ...	So that ...	Pri.
US-23	Finance Manager	be alerted to duplicate or unusual invoices with a risk score and reason	errors and fraud are caught early	S
US-24	Manager	ask the AI Assistant questions in natural language	I get answers from data I may see, with source and date	S
US-25	User	use the system in Arabic or English on desktop and mobile	it fits how and where I work	C

2.2 Use Cases

The use cases below cover the core demo scenario (budget to payment) and the selected AI features. Each requirement in Section 3 can be traced back to a use case or story.

UC-01 Create and Allocate a Budget	
Actors	CFO (primary); CEO (approves total)
Preconditions	User has budget-management permission; a financial period exists.
Main flow	1. CFO enters total budget and opening cash for the period. 2. CFO allocates amounts to departments, cost centers and expense categories. 3. System shows total, allocated and remaining amounts. 4. System validates the allocations and saves the budget.
Alternate / error flows	A. Allocations exceed the total: system rejects them unless a policy allows it with special approval. B. Negative or invalid amount: system rejects and explains why. C. Two users edit at once: system prevents the total from being exceeded.
Postconditions	Budget is saved with Planned amounts; every change is in the audit log with user, date, old and new values.

UC-02 Submit and Approve a Purchase Request	
Actors	Employee (maker); Department Manager and CFO (checkers)
Preconditions	Employee is logged in; department budget and approval rules exist.
Main flow	1. Employee fills the request: item, quantity, estimated price, department, category and justification. 2. System checks available budget and shows the result. 3. System routes the request by amount and approval rules. 4. Approver reviews and approves. 5. System reserves (commits) the amount and notifies the requester.
Alternate / error flows	A. Not enough budget: request is blocked or sent for special approval. B. Approver rejects or returns with a comment: requester is notified and budget is not reserved. C. Approver is the creator: system refuses (Maker–Checker).
Postconditions	Request is Approved, Committed amount is increased, and the timeline records every step.

UC-03 Issue Purchase Order and Receive Goods	
Actors	Procurement Officer; Warehouse Officer; Accountant
Preconditions	An approved purchase request exists; the supplier is registered.
Main flow	1. Procurement Officer creates a purchase order from the approved request. 2. Supplier delivers; Warehouse Officer records goods receipt and checks quantities. 3. System updates stock. 4. Accountant registers the supplier invoice. 5. System matches invoice, order and receipt.
Alternate / error flows	A. Partial delivery: order stays open for the rest. B. Quantity or price mismatch: system flags the difference and holds the invoice for review.
Postconditions	Stock is updated and a matched supplier invoice is ready for payment.

UC-04 Record a Supplier Payment	
Actors	Accountant (maker); Finance Manager (checker)
Preconditions	A matched supplier invoice exists; the payment is approved where policy requires.
Main flow	1. Accountant selects the invoice and chooses sandbox payment or external transfer. 2. For an external transfer, accountant enters reference and receipt. 3. System sets status Processing, then Paid. 4. In one atomic step the system updates cash balance, budget (Committed to Spent) and ledger.
Alternate / error flows	A. Payment fails: status Failed, invoice stays unpaid, reason recorded, retry allowed. B. Same confirmation arrives twice: system ignores the second one (idempotency key and unique reference). C. Missing reference: system rejects the record.
Postconditions	Invoice is Paid, cash, budget and ledger are consistent, and the action is in the audit log.

UC-05 Ask the AI Business Assistant	
Actors	Any user with data access
Preconditions	User is logged in; data exists in the system.
Main flow	1. User types a question, for example which department spent the most this month. 2. Assistant calls read-only tools that apply the user's own permissions. 3. Assistant returns an answer with the data source and date. 4. System logs the query.
Alternate / error flows	A. Data is outside the user's permissions: Assistant declines and does not reveal it. B. Not enough data: Assistant says so instead of guessing.
Postconditions	User gets an explainable answer; no data was changed.

UC-06 Review an Anomaly or Forecast Alert	
Actors	CFO; Accountant
Preconditions	Enough transactions exist (real or seed data); detection rules are configured.
Main flow	1. System checks new invoices and transactions against rules, then the model. 2. System creates an alert with a risk score and the reason. 3. User opens the alert and reviews the related records. 4. User confirms or dismisses it with a comment.
Alternate / error flows	A. Alert is a false positive: user dismisses it and the decision is recorded. B. Forecast shows a possible budget overrun: CFO is notified and may start a budget transfer request.
Postconditions	Alert has a recorded decision. The AI never pays, changes a budget or approves anything by itself.

3. Functional Requirements

Each requirement has a fixed ID (FR-module-number) that is carried into the SRD, the test cases and the traceability matrix. Priority: **M** Must, **S** Should, **C** Could. All Must items belong to Phase 1 (Core MVP).

3.1 Users, Roles and Access (AUT)

ID	Requirement	Pri.
FR-AUT-001	The system shall authenticate users with secure credentials and protect sessions (JWT or cookies).	M
FR-AUT-002	The system shall let an authorised user create users and assign roles and departments.	M
FR-AUT-003	The system shall check role permissions in the backend on every request, not only hide buttons in the interface.	M
FR-AUT-004	The system shall apply data scope (company, branch, department or own data) to every query.	M
FR-AUT-005	The system shall provide the default roles of the Blueprint and allow permissions to be customised.	S
FR-AUT-006	The system shall enforce Maker–Checker: the creator of a financial transaction shall not approve it, except for documented exceptions.	M
FR-AUT-007	The system shall let users change their password and manage their sessions.	S

3.2 Budget and Financial Planning (BUD)

ID	Requirement	Pri.
FR-BUD-001	The system shall let an authorised user create a budget for a financial period with total budget and opening cash.	M
FR-BUD-002	The system shall allocate the budget to departments, cost centers and expense categories, and shall not let allocations exceed the approved total unless a policy allows it with special approval.	M
FR-BUD-003	The system shall reject negative or invalid allocations and log every change with user, date, old and new values.	M
FR-BUD-004	The system shall prevent concurrent updates from exceeding the budget (race condition).	M
FR-BUD-005	The system shall show Planned, Committed, Spent and Available amounts for every allocation.	M
FR-BUD-006	The system shall allow limits per department or category and alert when a set percentage is reached.	S
FR-BUD-007	The system shall allow budget transfers between departments with the proper approval.	S
FR-BUD-008	The system shall compare planned against actual amounts.	S

3.3 Purchase Requests and Approvals (REQ)

ID	Requirement	Pri.
FR-REQ-001	The system shall let a user create a purchase request with item, quantity, estimated price, department, cost center, category, justification and attachments.	M
FR-REQ-002	The system shall check available budget when a request is submitted and show the result.	M
FR-REQ-003	The system shall route each request through an approval workflow based on amount and permissions.	M
FR-REQ-004	The system shall let an approver approve, reject or return a request with a comment, and record the decision.	M
FR-REQ-005	The system shall provide an approval inbox listing every item waiting for the user's decision.	M
FR-REQ-006	The system shall commit budget when a request is approved and release it when the request is rejected, cancelled or closed.	M
FR-REQ-007	The system shall show an activity timeline from creation to closure.	S
FR-REQ-008	The system shall allow approval delegation during absence.	C

3.4 Suppliers and Procurement (PRO)

ID	Requirement	Pri.
FR-PRO-001	The system shall manage suppliers and a catalog of products and materials.	M
FR-PRO-002	The system shall create a purchase order from an approved request for a chosen supplier.	M
FR-PRO-003	The system shall record goods receipt against a purchase order and check quantities, including partial deliveries.	M
FR-PRO-004	The system shall register supplier invoices linked to the order and receipt and flag mismatches.	M
FR-PRO-005	The system shall record quotations and compare suppliers.	C
FR-PRO-006	The system shall handle purchase order cancellation and returns.	C
FR-PRO-007	The system shall give suppliers limited access to their orders, invoices and payment status, without internal budgets.	C

3.5 Payments and Cash (PAY)

ID	Requirement	Pri.
FR-PAY-001	The system shall track each payment with the status Pending, Processing, Paid, Failed or Refunded.	M
FR-PAY-002	The system shall support a sandbox payment simulation.	M
FR-PAY-003	The system shall record external transfers with reference number, date and receipt.	M
FR-PAY-004	The system shall process a repeated payment confirmation only once, using an idempotency key and a unique reference constraint.	M
FR-PAY-005	The system shall keep the invoice unpaid when a payment fails, record the reason and allow a retry.	M
FR-PAY-006	The system shall, in one atomic transaction, update cash balance, budget (Committed to Spent) and ledger when a payment succeeds.	M
FR-PAY-007	The system shall manage cash accounts and balances.	M
FR-PAY-008	The system shall not move real money in the first version; real integration only after the provider officially confirms support and legal requirements are checked.	M

3.6 Accounting (ACC)

ID	Requirement	Pri.
FR-ACC-001	The system shall provide a basic chart of accounts.	M
FR-ACC-002	The system shall create balanced double-entry ledger entries automatically from financial transactions.	M
FR-ACC-003	The system shall not allow posted entries to be silently edited or deleted; corrections shall use an approved adjusting entry.	M
FR-ACC-004	The system shall provide basic reports (budget vs actual, expenses by department and category, ledger) with PDF or Excel export by permission.	S
FR-ACC-005	The system shall support period closing.	C

3.7 Inventory (INV)

ID	Requirement	Pri.
FR-INV-001	The system shall manage items, units of measure and warehouses.	M
FR-INV-002	The system shall increase stock when goods receipt is recorded.	M
FR-INV-003	The system shall record stock issue and transfers between warehouses.	S
FR-INV-004	The system shall alert when stock falls below the minimum level.	S
FR-INV-005	The system shall support stocktaking and stock adjustments.	C

3.8 Dashboards, Notifications and Audit (SYS)

ID	Requirement	Pri.
FR-SYS-001	The system shall show role-based dashboards (CEO/CFO, Department Manager, Employee, Warehouse).	M
FR-SYS-002	The system shall record every sensitive action in an audit log (who, when, old and new value) that cannot be silently deleted.	M
FR-SYS-003	The system shall send notifications for pending approvals, failed payments, low stock and budget thresholds.	S
FR-SYS-004	The system shall allow attachments on requests, invoices and receipts.	S
FR-SYS-005	The system shall provide filters by date, department, supplier and status, with exports by permission.	S
FR-SYS-006	The system shall provide global search limited by permissions, and internal comments on requests.	C

3.9 AI Features (AI)

ID	Requirement	Pri.
FR-AI-001	The system shall forecast cash flow for the next week, month or quarter and evaluate it against a baseline.	S
FR-AI-002	The system shall estimate the probability that a department will exceed its budget.	S
FR-AI-003	The system shall detect anomalies (duplicate invoices, unusual prices or quantities, invoice–order–receipt differences) using rules first, then a model such as Isolation Forest, and give a risk score with the reason.	S
FR-AI-004	The AI Assistant shall answer natural-language questions using only permission-checked read tools, show the data source and date, and log each query.	S
FR-AI-005	The AI shall only suggest actions; it shall never pay, change a budget or approve a transaction by itself.	M
FR-AI-006	The system shall store AI recommendations and show the data source and date with every AI result.	S

Count: 60 functional requirements. Phase 2–4 features (sales, payroll, production, bank reconciliation, multi-company, multi-currency, workflow builder) are out of this list and stay as future work.

4. Non-functional Requirements

Targets come from the KPIs in the Project Planning & Management document. Values marked **To be set in SRD** need the expected system size (users, data volume) before a number can be fixed.

ID	Category	Requirement	Target / Measurement
NFR-SEC-01	Security	Authentication, session protection and encrypted connections; RBAC and data scope checked in the backend on every request	100% unauthorized access blocked in role-based tests
NFR-SEC-02	Privacy	Sensitive data (salaries, financial details, customer data) is visible only with explicit permission, also through the AI Assistant	100% in permission tests per role
NFR-INT-01	Financial integrity	Atomic transactions, no duplicate payments, accurate budget and ledger figures, balanced entries	100% in reconciliation tests
NFR-AUD-01	Auditability	Sensitive actions are logged with user, time, old and new values; history cannot be silently deleted	100% of sensitive actions in audit log review
NFR-REL-01	Reliability	Correct handling of payment failure, connection loss and retries; no half-completed postings	Failure and retry test cases pass
NFR-REL-02	Backup and recovery	Regular backups with a tested restore; database scripts and seed data under version control	RTO / RPO: To be set in SRD
NFR-PER-01	Performance	API calls respond quickly under the expected load	Under 2 seconds (API tests and logs); final numbers per data size in SRD
NFR-PER-02	Performance	Dashboards load quickly	Under 3 seconds (browser tools)
NFR-AVL-01	Availability	System is available during working and demo hours	To be set in SRD
NFR-SCA-01	Scalability	Modular monolith design that allows more users, branches and data later	Clear module boundaries (Finance, Procurement, Inventory)
NFR-USA-01	Usability	Clear screens, understandable error messages, role-specific dashboards, easy filters and reports	At least 4 of 5 satisfaction; at least 90% task success
NFR-USA-02	Usability	Responsive interface for desktop and mobile; Arabic and English with suitable date and currency display	Main flows usable on both screen sizes; language support is Should
NFR-MNT-01	Maintainability	Separated modules, documented APIs (OpenAPI), automated tests, code reviews on every pull request	Unit coverage at least 70% on financial and permission logic; 100% PRs reviewed
NFR-TST-01	Testability / Quality	Requirements are covered by test cases and tests run in CI	At least 90% requirements covered; pass rate at least 95%; 0 critical bugs at delivery
NFR-AIG-01	AI governance	AI results are explainable (source and date), recommendations are logged, AI respects user permissions and always keeps human approval	100% in permission tests; MAPE for forecasting; precision and recall for anomalies

5. Assumptions & Open Items

5.1 Assumptions

- NUVANTA is designed as a general, customizable system for one company with several branches and departments (to be confirmed in the SRD).
- The first version uses sandbox payments and external-transfer recording; no real money moves.
- AI features use realistic seed or synthetic data, starting with rule-based detection.
- Stakeholder needs come from the Blueprint and domain knowledge, and will be refined with real user feedback when available.

5.2 Open Items for the SRD

#	Open Item	Why it matters
1	Type of company used for the workflows (manufacturing, trading, services or general)	Decides which Phase 2 modules and sample data are needed
2	Single company or multi-company (SaaS)	Changes data isolation and the database design
3	Expected number of users and data volume	Needed to fix performance, availability and RTO / RPO numbers
4	Approval limits and exception policy for Maker–Checker	Defines approval rules and permission tests
5	Whether Arabic/English and the supplier portal are included in the first version	Currently ranked Should / Could